Vaidehi Pujary

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Education

Purdue University West Lafayette, IN
Master of Science in Electrical and Computer Engineering (Thesis Track) May 2027 (Expected)

The University of Arizona Tucson, AZ
Bachelor of Science in Electrical and Computer Engineering May 2025
Minor in Optical Sciences and Engineering GPA: 4.0/4.0 (Honors)
Minor in Mathematics

- **Relevant Coursework**: Digital Cameras and Fourier Optics, Semiconductor Physics and Lasers, Computer Programming for Engineering Applications, Object-Oriented Software Design, Computer Architecture, Microprocessor Organization, Electrical and Optical Properties of Materials, Digital Signal Processing, Circuit Theory, Semiconductor Processing, Linear Algebra, Probability and Statistics
- **Activities and Societies**: Baja Wildcat Racing, K7UAZ Amateur Radio (Technician Class License, KK7PKR), Society of Women Engineers (Outreach Committee), Tau Beta Pi, Campus Pantry Volunteer
- **Honors and Awards**: Academic Year Highest Academic Distinction, Spirit of Inquiry Award, Grace Hopper Conference Scholar, W.L. Gore & Associates All in the Same Boat Award, Global Wildcat Tuition Award

Technical Skills

Languages: Python, C#/.NET, Java, C, C++, Verilog, Assembly, JavaScript, HTML/CSS, MATLAB, SQL
Libraries and Tools: Git, NumPy, Matplotlib, SciPy, FPGA, Arduino
Software: Xilinx Vivado, Cadence PSpice, Code V, Zemax OpticStudio, PyCharm, Ubuntu, Robot Operating System

Work Experience

Graduate Teaching Assistant August 2025 – Present
Elmore Family School of Electrical and Computer Engineering, Purdue University *West Lafayette, IN*

- Lead weekly lab sessions for 80+ Fundamentals of Electrical Engineering students, delivering demonstrations and supervising hands-on experiments involving oscilloscopes, function generators, multimeters, and power supplies.
- Guide students through systematic troubleshooting of circuits and instrumentation, diagnosing wiring errors, instrument setup issues, and measurement inaccuracies.
- Evaluate weekly assignments, formal reports, laboratory practicals, and demonstration projects.

Optical Engineering Intern June 2025 – August 2025
4D Technology Corporation, an Onto Innovation Subsidiary *Tucson, AZ*

- Developed Python simulations, synthetic data, and data processing pipelines to model optical system performance, supporting design feasibility assessments for R&D.
- Executed test plans and took measurements using new optical metrology systems and proof-of-concept prototypes, contributing to accelerated product development cycles and data-driven design decisions.
- Assembled optical breadboard systems and conducted quality control tests, ensuring accuracy and reliability of instruments.

Undergraduate Research Assistant March 2022 – May 2025
Model Based Design Laboratory, University of Arizona *Tucson, AZ*

- Developed Arduino Nano software to process sensor data, enabling the detection of the extent of tissue deformation in robotic surgery.
- Wrote a Python data analysis script and assisted in developing a ROS node application to provide real-time visualization of sensor data.

- Contributed to the formulation of a fuzzy classification algorithm designed to actuate vibration motors, utilizing processed sensor data to map pressure sensations to precise frequency and amplitude control for vibrations.
- Evaluated the system's feasibility by conducting over 100 experimental trials using objects with varying material properties.
- Designed and implemented a PCB to optimize signal integrity and improve hardware portability, and established RS232 communication between Linux and Windows systems for seamless data exchange.

Application Developer January 2023 – May 2025
Eller Systems Group, Eller College of Management *Tucson, AZ*

- Collaborated with a team of 6 to develop, test, and debug a C# .NET student management web application used by 6000+ people.
- Implemented user-requested features into the MVC application while adhering to the constraints of a monthly release cycle.
- Developed key features for custom remote learning software, such as live chat and SignalR hand raise functionality, based on client requirements and specifications.

R&D Electrical Engineering Intern May 2024 – August 2024
Cepheid *Sunnyvale, CA*

- Spearheaded the design of an experimental setup to characterize optoelectronic components, developed multiple test procedures, and collected data from 200+ trials to support design verification activities.
- Automated the test fixture using Python scripts and built a custom graphical user interface for user input, reducing test time by 94% and improving the efficiency of testing processes for quicker data collection and analysis.
- Diagnosed root causes of failures in test assembly and implemented corrective actions, while creating technical documentation for operating procedures.

Undergraduate Lab Assistant January 2022 – May 2022
Electrical and Computer Engineering Department, University of Arizona *Tucson, AZ*

- Guided 90+ Introduction to C students in debugging lab programs coded in C, offering insightful suggestions for code optimization.
- Facilitated weekly lab sessions, addressing student queries concerning lecture topics and fostering a deeper understanding of the subject matter.
- Evaluated and provided feedback on laboratory assignments, examinations, and homework submissions.

Projects

ASML Optical Metrology Module | Senior Design Project August 2024 - May 2025

- Designed and implemented a Python-based real-time particle tracking GUI software capable of characterizing particle size, velocity, and trajectory with sub-pixel accuracy using image processing techniques.
- Performed optical lab testing to verify system requirements and communicated experimental findings, assisting in the refinement of optical setups and data processing methods.
- Coordinated with cross-functional team to rapidly iterate on hardware/software integration, and authored technical documentation for future development.

MIPS Processor on FPGA | Computer Architecture Course August 2023 – December 2023

- Designed and implemented a 5-stage pipelined processor for the MIPS-32 ISA on a Xilinx Artix-7 FPGA using Verilog and Vivado in a team of 3, utilizing forwarding to increase throughput and reduce stalls.
- Optimized a video compression algorithm in assembly by introducing custom instructions.
- Conducted post-synthesis and post-routing functional and timing verification through Verilog testbench simulations and timing reports before successfully running the design on an FPGA.